

01. TITLE SLIDE

~1 min

CONTENT SUMMARY

Session 11: Advanced Architectures • Subtitle: Cutting-Edge AI Models

Welcome to Session 11. We've now covered the complete AI engineering lifecycle, and mastered safety and fairness. Now: explore the cutting edge. Advanced architectures that power modern AI systems. Transformers, attention mechanisms, mixture of experts, multi-modal learning. These aren't theoretical. They're how GPT-4, Claude, Gemini, and other state-of-the-art models work. By end of today, you'll understand the architecture of modern foundation models. You'll know how attention works, why mixture of experts matters, how multi-modal learning combines different modalities, and how companies build next-generation AI. This is the frontier of AI.

02. TODAY'S JOURNEY

~2 min

CONTENT SUMMARY

14 slides covering: architectures, attention, experts, multi-modal

Quick roadmap. Five minutes on why advanced architectures matter. Eight minutes on transformer deep dive—the foundation of modern LLMs. Eight minutes on attention mechanisms. Eight minutes on mixture of experts (scaling to billions of parameters). Eight minutes on multi-modal learning (text + images + audio). Eight minutes on Vision Transformers. Five minutes on cross-attention and fusion. Five minutes on parameter efficiency. Eight minutes on real-world architectures. Five minutes on choosing architectures. Lab and takeaways. Dense session. Protect transformers, attention, multi-modal. Those are core.

03. WHY ADVANCED ARCHITECTURES MATTER

~5 min

CONTENT SUMMARY

Performance, efficiency, capabilities, scale

Advanced architectures enable better models. GPT-3 (175B params) to GPT-4 (unknown, but estimated 100T+ with mixture of experts). Scaling requires architectural innovation. Transformers: 100x better than RNNs at parallelization. Attention: Focus on relevant context. Mixture of experts: Scale to trillions of parameters efficiently. Multi-modal: Combine text, images, audio for richer understanding. Vision Transformers: Beat CNNs at image understanding. Real impact: GPT-3 to GPT-4 improvement (45% to 86% accuracy on exams). That's architectural innovation. Foundation models rest on these architectures. If you understand architecture, you understand modern AI.

04. TRANSFORMER DEEP DIVE

~8 min

CONTENT SUMMARY

Encoder-decoder, self-attention, positional encoding, scale

Transformer (2017): "Attention is All You Need" paper revolutionized NLP. Before: RNNs, LSTMs (sequential, slow to train, hard to parallelize). After: Transformers (parallel, fast, scales. Architecture: (1) Embedding layer: Convert tokens to vectors. (2) Positional encoding: Add position information (transformers don't inherently know word order). (3) Self-attention: Each token attends to all other tokens, learns which are important. (4) Feed-forward: Dense layers. (5) Layer norm: Normalization. Repeated 12-96 times (12 for BERT, 96 for GPT-3). Encoder vs decoder: Encoder (BERT style): Bidirectional, attends to left and right. Decoder (GPT style): Unidirectional, attends only to left (previous tokens). Encoder-decoder (T5, seq2seq): Encoder understands input, decoder generates output. Why transformers work: Parallelization (train on GPUs efficiently), long-range context (attention can connect far tokens), scaling laws (add more parameters, more data, get better). GPT-3: 175B parameters, 300B tokens of training data, 2-3 trillion FLOPs. Modern scaling: GPT-4 estimated 1.7 trillion parameters (with mixture of experts making it efficient).

05. ATTENTION MECHANISMS

~8 min

CONTENT SUMMARY

Self-attention, multi-head, cross-attention, scaled dot-product

Attention: For each token, compute relevance to all other tokens. Scaled dot-product attention: Query Q, Key K, Value V. $\text{Score} = \text{softmax}(Q K^T / \sqrt{d})$. Output = Score $\times$ V. Intuition: For each token, compute how much to attend to each other token. Scale by $\sqrt{d}$ for stability. Multi-head attention: Multiple attention heads in parallel. Head 1: Attends to syntax. Head 2: Attends to semantics. Head 3: Long-range dependencies. Heads learn different aspects. 8 heads in BERT, 96 heads in GPT-3. Each head: 64 dimensions (smaller, efficient). Cross-attention: Query from one sequence, Key/Value from another. Example: Generating image caption. Image tokens (encoder) + previous words (decoder) attend to image. Key innovation: Enables fusion of different modalities. Why attention matters: Long-range dependencies (CNN sees only local context, attention sees everything). Interpretability (can visualize which tokens attended to which). Scaling (parallelizable). Attention is why transformers work.

06. MIXTURE OF EXPERTS

~8 min

CONTENT SUMMARY

Conditional computation, sparse models, scaling, routing

Problem: 175B parameters = massive. More parameters = better performance. But scale to 1 trillion params? Impractical. Solution: Mixture of experts. Sparse model: Not all parameters used for all inputs. Router network: For each token, decides which experts to use. Experts: Specialized neural networks. Example: Expert 1 good at pronouns, Expert 2 good at technical terms, Expert 3 good at verb conjugation. Token routes to 1-2 relevant experts. Only those experts' parameters compute for that token. Result: Scale to trillions of parameters efficiently. Google: Switch Transformer with 1.6 trillion parameters. Trained faster than dense 180B model, better performance. How it works: Input token → Router (learned) → Selects top-K experts (e.g., top 2) → Those experts compute → Output combines results. Efficiency: 1.6 trillion parameters, but uses only fraction per token. Training: Challenge is load balancing (experts should be used equally). Solution: Add loss term penalizing imbalance. Real deployment: GPT-4 rumored to use mixture of 8 experts, 220B params each. Only 2 active per token. Effective: 1.76 trillion params, but ~440B active.

07. MULTI-MODAL LEARNING

~8 min

CONTENT SUMMARY

Text + images + audio, joint embeddings, alignment

Multi-modal: Combine multiple modalities (text, images, audio, video). CLIP: Text-image alignment. Train on 400M image-text pairs. Image encoder (ViT) and text encoder (BERT) learn shared embedding space. Similar image and caption have similar embeddings. Result: Can search images by text, understand relationships. GPT-4V: Multimodal GPT-4. Takes images and text, understands both. Reads tables, charts, handwritten text in images. Alignment: Key challenge: How to combine modalities? (1) Early fusion: Concatenate embeddings. Problem: Different modalities have different scales. (2) Cross-modal attention: Image tokens attend to text, text attends to image. Each modality refines understanding of other. (3) Separate encoders: Each modality has encoder, fuse at higher level. CLIP approach: Separate text and image encoders, align in embedding space. Real impact: CLIP enables zero-shot image classification. Show image. Align with text descriptions of classes. Highest similarity wins. No training needed. Applications: Image search, content moderation (describe policy, search for violations), accessibility (describe images for blind users).

08. VISION TRANSFORMERS

~8 min

CONTENT SUMMARY

Image as patches, positional embeddings, image understanding

Vision Transformers (ViT): Apply transformer architecture to images. Before: CNNs (local receptive field, need many layers for global context). After: ViT (global context from start). How: (1) Split image into 16x16 patches (e.g., 224x224 image → 196 patches). (2) Embed each patch (to vector space). (3) Add positional embeddings (which patch is where). (4) Pass through transformer. (5) Use [CLS] token (first token) for classification. Result: ViT-L (300M params) outperforms ResNet-152 (60M params) on ImageNet. Better accuracy, better scaling. Why: Attention allows long-range pixel dependencies. CNN needs many layers. Multi-head attention: Different heads focus on different features. Head 1: Edge detection. Head 2: Texture. Head 3: Object shape. Scaling: More data, larger models → ViT scales better than CNNs. Real deployment: Google uses ViT for image understanding. Meta uses ViT for image encoders in multi-modal systems.

09. CROSS-ATTENTION & FUSION

~5 min

CONTENT SUMMARY

Combining modalities, attention between sequences, practical fusion

Cross-attention: Different sequences attend to each other. Example: Image caption generation. Image tokens (from image encoder) as Key/Value. Previous words (from text decoder) as Query. Decoder attends to image to decide next word. Practical: For hiring AI with multi-modal input (resume text + profile photo). Text encoder processes resume. Image encoder processes photo. Cross-attention: Resume attends to photo, photo context informs resume understanding. Fusion strategies: (1) Concatenation: Simple but naive. (2) Addition: Element-wise addition of embeddings. (3) Multiplication: Element-wise product. (4) Attention-based: One modality attends to other. (5) Learned fusion: Neural network learns how to combine. Best: Attention-based or learned fusion (adaptive to data). Real example: BERT-ViT: BERT for text, ViT for images, cross-attention between them. Used for visual question answering (given image and question, answer about image).

10. PARAMETER EFFICIENCY

~5 min

CONTENT SUMMARY

LoRA, adapter modules, knowledge distillation, quantization

Challenge: Foundation models huge. GPT-3: 175B params. GPT-4: 1.7T params (estimated). Can't fine-tune on edge devices or small budgets. Solutions: LoRA (Low-Rank Adaptation): Instead of fine-tuning all params, add small trainable matrices. For each weight matrix W , add $\Delta W = A \times B$ (low-rank). Train only A and B (much smaller). Result: 10K-100K trainable params instead of billions. Accuracy nearly same as full fine-tuning. Adapters: Insert small modules between layers. Train only adapters. Rest of model frozen. Prefix tuning: Add learnable prefix to input. Model processes prefix + input. Only prefix trainable. Quantization: 8-bit or 4-bit weights instead of 32-bit. 4x size reduction, 4x faster inference. Combines with distillation: Smaller teacher model, distill to even smaller student. Real deployment: LLaMA-LoRA: Take 13B parameter LLaMA, add LoRA modules, fine-tune on single GPU. Previously required 8-GPU setup. Results nearly same.

11. REAL-WORLD ADVANCED ARCHITECTURES

~8 min

CONTENT SUMMARY

GPT-4, Claude, Gemini, Llama, Mistral architectures

GPT-4: Mixture of experts (estimated 8 experts, ~220B each). Only 2 active per token. Mixture of modalities (text and vision). Advanced attention mechanisms. Custom hardware (TPU-4 clusters). Cost: Billions to train. Claude: Transformer with advanced attention (multi-query attention instead of multi-head for efficiency). Constitutional AI training (follow values). Long context (100K tokens). Gemini: Multi-modal from ground up. Text, image, video, audio inputs. Trained on 6M+ H100 GPUs. Cross-modal attention throughout. Llama 2: Open source. 7B-70B parameter models. Focused on efficiency. Good for edge deployment. Mistral: Sparse Mixture of Experts. 8x7B experts, 2 active per token. Smaller than Llama but more capable. Real comparison: Llama 70B: 70B params, ~2 weeks training on 2000 GPUs. Gemini Ultra: Unknown params, estimated 1-2 trillion, months of training. Performance gap: GPT-4 > Claude > Gemini Pro > Llama 70B > Mistral 8x7B. But tradeoff: Smaller models more efficient. Deployment: Cloud (GPT-4), Edge (Mistral), On-device (LoRA Llama).

12. CHOOSING ARCHITECTURES

~5 min

CONTENT SUMMARY

Decision framework, tradeoffs, deployment constraints

Choosing architecture depends on constraints. Question 1: Compute budget? (1) Unlimited (billions): Go for GPT-4 scale. (2) Millions: Use Llama or Mistral. (3) Thousands: Use small LoRA model. (4) Limited (laptop): Use 7B model + quantization. Question 2: Latency? (1) Real-time required (<100ms): Use smaller model, quantization, mixture of experts. (2) Batch processing okay (seconds): Larger model okay. Question 3: Multi-modal? (1) Yes: Use architecture with cross-attention, vision transformer. (2) No: Pure language model fine. Question 4: Fine-tuning needed? (1) Yes, custom domain: Use adapter modules or LoRA. (2) No, use as-is: Base model fine. Decision matrix: Most projects: Start with Llama 7B, fine-tune with LoRA, quantize to INT8, deploy on edge. Works for most cases, reasonable cost. If advanced multi-modal needed: Add vision transformer, cross-attention. If scale matters: Mixture of experts approach. If open source critical: Llama or Mistral. If best performance needed: Proprietary (GPT-4, Claude).

13. LAB EXERCISE OVERVIEW

~3 min

CONTENT SUMMARY

Design advanced architecture for hiring AI

Your task: Design an advanced architecture for your hiring AI. What architectural choices? Multi-modal (resume text + photo)? Mixture of experts? Advanced attention? Why? Lab has 6 parts: (1) Architecture choice (2 min), (2) Design rationale (2 min), (3) Data flow (1 min), (4) Parameter efficiency analysis (2 min), (5) Compute cost estimate (2 min), (6) Comparison to simpler approach (2 min). Deliverables: Architecture diagram, design rationale, data flow, efficiency analysis, cost breakdown, comparison.

14. KEY TAKEAWAYS FROM SESSION 11

~5 min

CONTENT SUMMARY

6 Insights: Transformers revolutionized, Attention is powerful, Mixture experts scale, Multi-modal future, Efficiency critical, Right architecture matters

Takeaway One: Transformers revolutionized NLP. Parallelizable, scalable, powerful. Takeaway Two: Attention mechanisms enable long-range dependencies and interpretability. Takeaway Three: Mixture of experts scales to trillions efficiently. Takeaway Four: Multi-modal learning combines modalities. Future is multi-modal. Takeaway Five: Parameter efficiency critical for deployment. LoRA, adapters, quantization. Takeaway Six: Right architecture depends on constraints. No one-size-fits-all.

15. COMING UP NEXT

~2 min

CONTENT SUMMARY

Session 12: Real-World Case Studies & Capstone

Session 12: Capstone project. Build end-to-end AI system using everything from Sessions 1-11. See you there.